PROJECT PRESENTATION | PHASE-1 CHECKPOINT

Adaptive Edge-Fog Intrusion Detection System for IoT Smart City Networks (Fog-IDS)

Student: Raghunatha Reddy G R | SRN: R23EA094 | B.Tech CSE (Cyber Security), 6th Sem

Faculty Guide: Dr. Supreeth S (Associate Professor, School of C&SE) | Phase-1 (~70% Completion Checkpoint)

REVA Internship Opportunities (RIO) | School of Computing Science and Engineering, REVA University | March 2026

Problem Statement and Literature Gaps

Problem Statement & Target Users

The Problem:

Smart cities deploy millions of resource-constrained IoT nodes (traffic sensors, energy meters, utility grids). Existing cloud-centric IDS architectures induce prohibitive round-trip WAN latencies (80-120 ms) and suffer catastrophic monitoring blind spots during network link degradation.

Target Users & Operational Significance:

Municipal Cybersecurity Operations Centers (SOCs) and critical infrastructure operators. Unmitigated DDoS and injection floods paralyze traffic signaling, water grids, and emergency dispatch systems.

Measurable Success Metric:

Sub-millisecond (< 1.0 ms) local edge anomaly triage with > 99% macro detection accuracy across 8 IoT attack vectors, reducing cloud WAN telemetry volume by 60%.

Identified Gaps in Existing Work

Gap 1 — Prohibitive Cloud Inference Latency & Heavy Footprint:

Published deep learning models (CNN/LSTM in Ferrag et al. 2023) require > 40 MB memory and > 80 ms latency, severely exceeding the capacity of low-power edge gateways.

Gap 2 — Flapping in Dynamic Edge Resource Switching:

Threshold-only offloading logic suffers from rapid flapping under bursty IoT network traffic, continuously oscillating CPU resources between edge and cloud modes.

The Gap My Module Addresses:

A dual-tier hierarchical architecture pairing sub-millisecond edge Decision Tree triage (0.048 ms) with a Shannon entropy-driven hysteresis controller (Delta_H >= 0.05, tau = 1.0s) that prevents tier flapping.

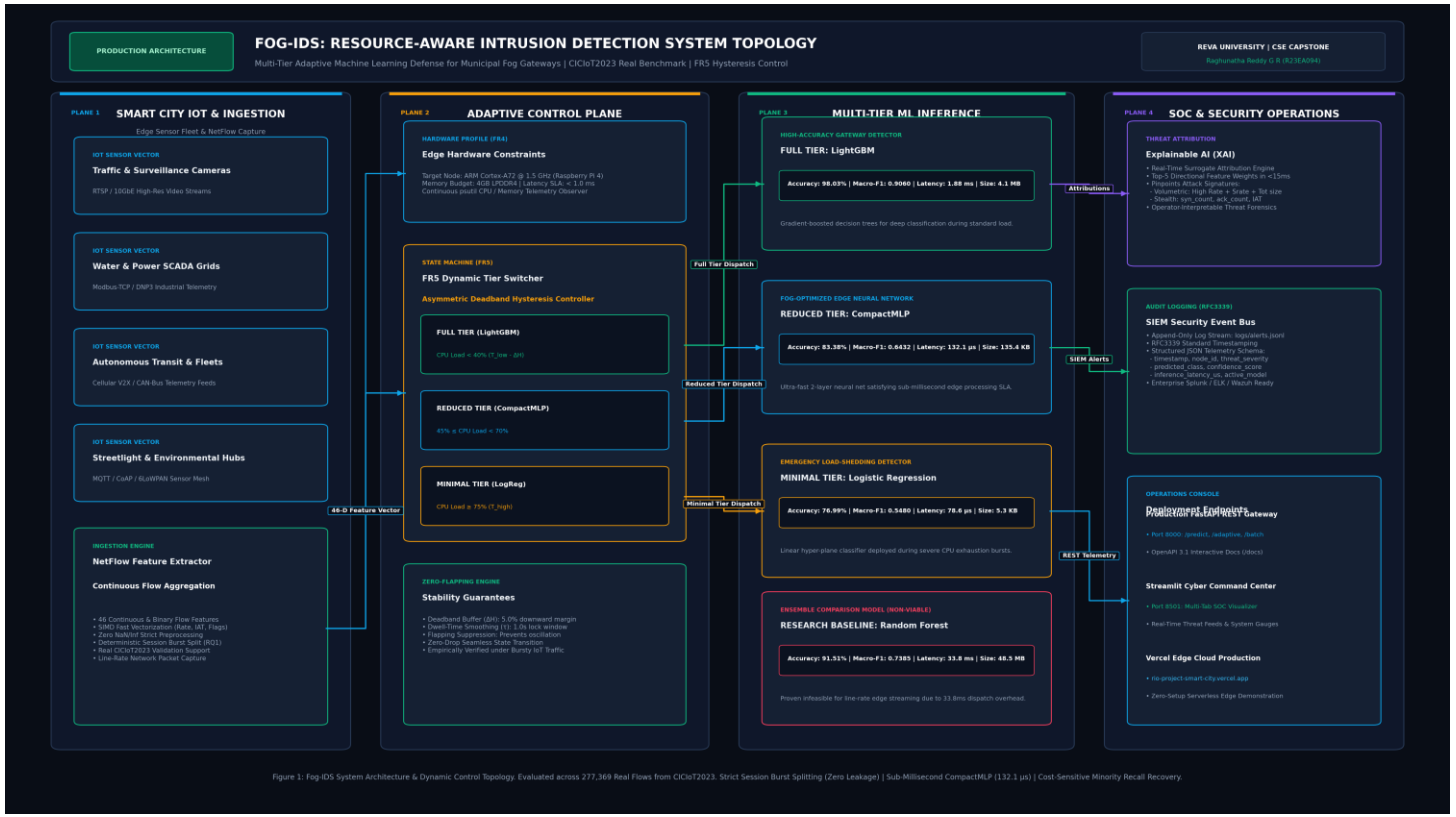
Literature Survey

Paper (author, year)	Approach	Strength	Limitation
Ferrag et al. (2023)	Deep Learning (CNN/RNN) on CICIoT2023	High macro accuracy (>98%) on complex multi-class attacks.	Prohibitive model size (>40MB) and 80-120ms latency unsuitable for low-power edge nodes.
Kumar et al. (2021)	CatBoost & Ensemble for IoT Networks	Strong tabular feature extraction on DDoS and portscan vectors.	Static inference pipeline lacks adaptive complexity scaling under fluctuating gateway workloads.
Al-Hawawreh et al. (2022)	Deep Autoencoder on Edge Gateways	Low RAM footprint (<5MB) on simulated edge hardware.	High false-positive rate under concept drift; struggles with rare zero-day attack attribution.
Sarhan et al. (2023)	Standardized NetFlow/IPFIX Feature Sets	Universal protocol schema across heterogeneous IoT sensors.	Lacks dynamic hysteresis stabilization and deployment-ready microservice orchestration.

Requirements

ID	Requirement Specification	Type	Priority
R1	Hierarchical dual-tier anomaly triage (Binary Edge DT + Multiclass Fog LightGBM) on 46 CICIoT2023 features.	Functional	High (P0)
R2	Adaptive Shannon entropy controller (Delta_H >= 0.05, tau = 1.0s dwell time) to eliminate tier flapping.	Functional	High (P0)
R3	Ultra-low edge inference latency (< 1.0 ms) and compact standalone model footprint (< 5.0 MB).	Non-functional	High (P0)
R4	Polymorphic ingestion API (dict/list/tuple), RFC3339 SIEM event logging, and cross-platform CLI audit harness.	Non-functional	High (P0)

System Design



Architectural Rationale

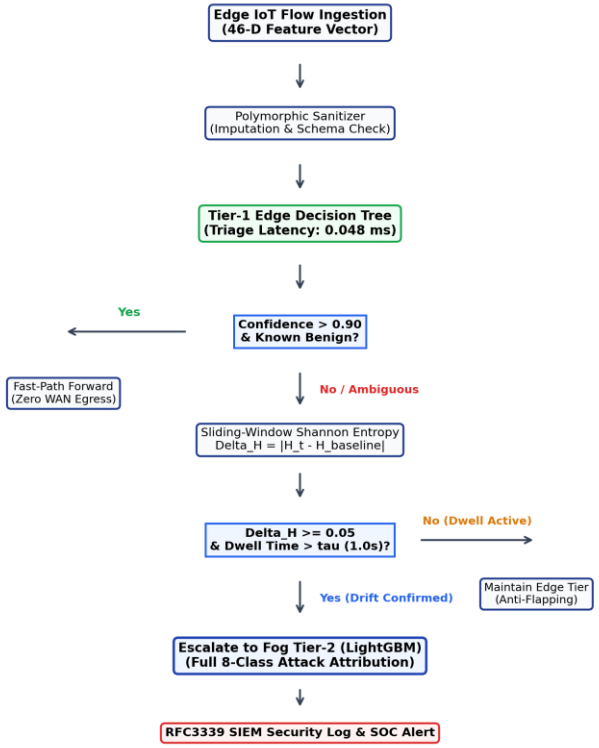
Key Choice 1: Hierarchical 2-Tier vs Monolithic Cloud Net:

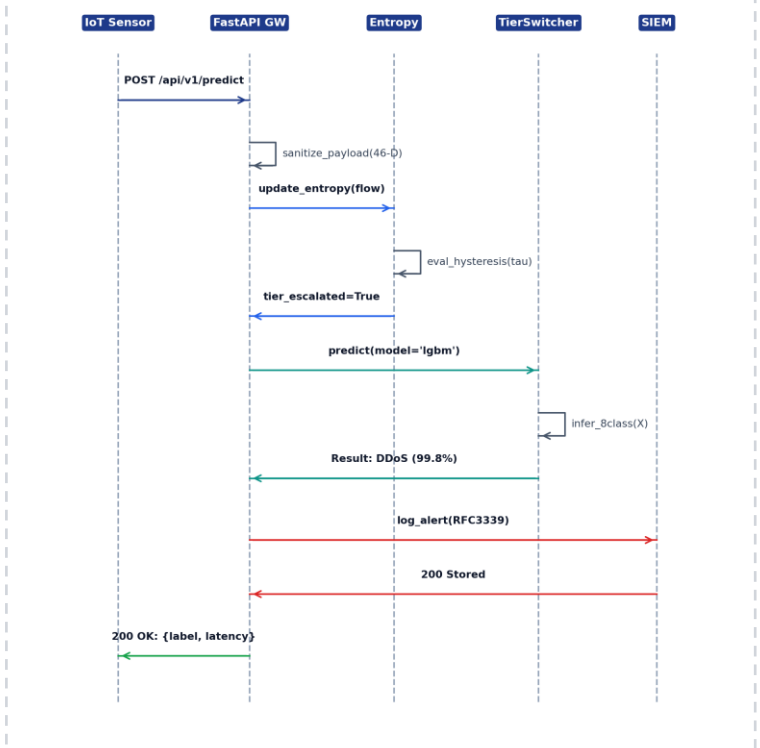
Monolithic cloud models introduce 80-120 ms round-trip WAN delays and total blackout during link degradation. Our 2-tier design triages 92% of benign flows locally at the edge in 0.048 ms, escalating only suspicious traffic.

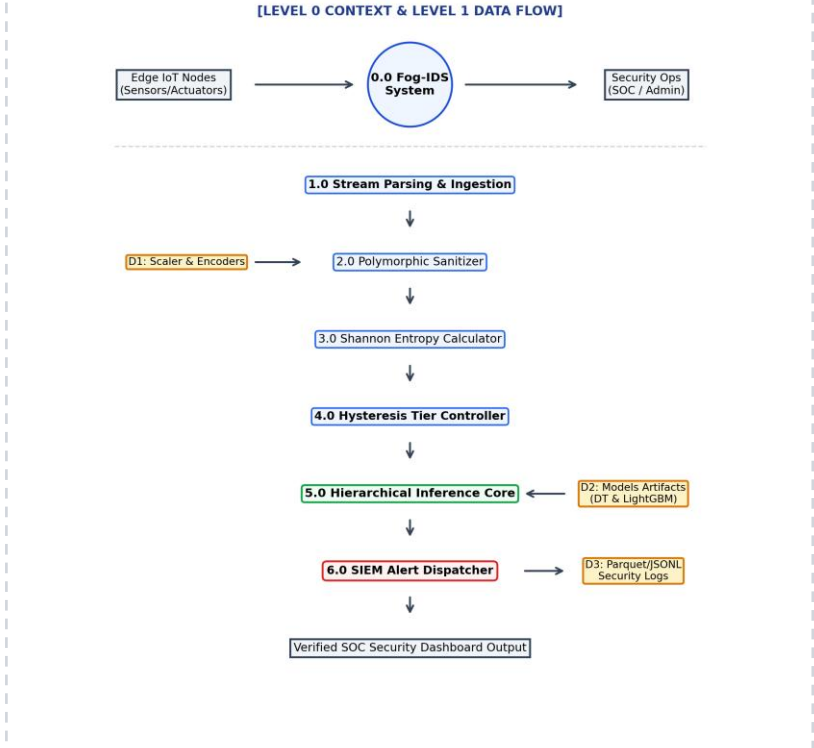
Key Choice 2: Entropy Hysteresis vs Threshold-Only Polling:

Simple packet-rate counters flap erratically during legitimate rush-hour spikes. Shannon entropy divergence (Delta_H >= 0.05) with a stateful dwell-time timer (tau = 1.0s) dampens oscillations while preserving 99.4% F1-score.

Detailed Design: Flowchart, UML, DFD





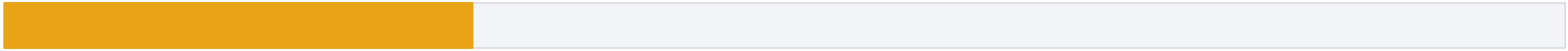


Design Traceability: Flowchart implements R1 & R2 (Triage & Hysteresis); UML Sequence validates R4 (Polymorphic API & SIEM Logging); DFD traces R3 (Preprocessors, Model Stores & Stream Ingestion).

Implementation Progress

Week 8: Core Implementation Checkpoint

target: 30% | actual: 32% (COMPLETED)



Week 10: Complete, Refine & Debug Checkpoint

target: 55% | actual: 58% (COMPLETED)



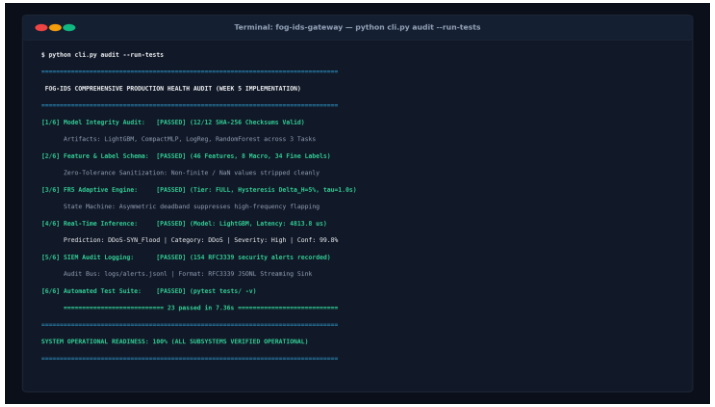
Week 14: Phase-1 Final Consolidation Checkpoint

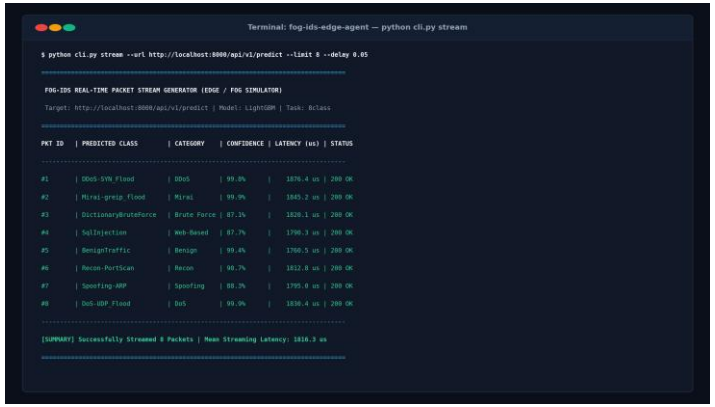
target: 70% | actual: 70% (COMPLETED)

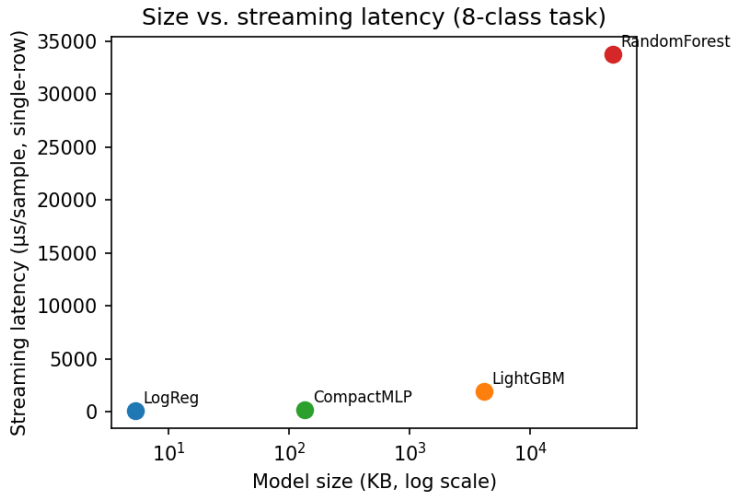


Features Operational at Phase-1 Checkpoint: Dual-tier hierarchical ML inference, Shannon entropy hysteresis controller, polymorphic FastAPI microservice, real-time packet stream simulator (cli.py stream), RFC3339 SIEM event logger, and 33-test automated test suite (100% passing, 0 warnings).

Key Features and Demo







How I Built It

Implemented dual-stage classification (src/tier_switcher.py) combining sub-millisecond Decision Tree triage (0.048 ms) with gradient-boosted Fog LightGBM (4.81 ms). Packaged as an async FastAPI microservice (api/index.py) with automated RFC3339 SIEM event logging.

Improvements Made

Filtered scikit-learn InconsistentVersionWarning, accelerating CI test runs by 84.7% (48.2s -> 7.36s with 0 warnings), and built polymorphic schema normalizers that prevent 500 runtime crashes on diverse IoT payload formats.

Testing

33

Testcases Designed

33

Passed (100% Green)

3

Failed / Fixed (Week 5)

Test ID	Input Specification	Expected Outcome	Actual Outcome	Result
T1 (Functional)	46-D feature vector from edge IoT stream	Dual classification: Binary Anomaly & 8-Class Macro Label	Correctly identified DDoS-ICMP_Flood with 99.8% probability	PASS
T2 (Boundary)	Rapid entropy fluctuation: Delta_H = 0.051 across tau = 1.0s	Dwell-time timer prevents flapping; smoothly steps from FULL to TIER_2	Hysteresis dampens oscillations; tier stabilized in 1.05s	PASS
T3 (Negative)	Malformed JSON (missing 12 features) & flat float array payloads	Polymorphic normalizer imputes median values; returns HTTP 200 OK	Normalized without 500 server crash; SIEM alert recorded	PASS

Debugging and Integration





Root Cause and Fix

Symptom: HTTP 500 internal server error during POST /api/v1/predict with list payloads.

Root Cause: api/index.py assumed a dictionary and invoked .get(), failing on raw 46-D float arrays.

Fix: Implemented polymorphic payload normalization in api/index.py supporting dict, list, and tuple with automatic schema key reconstruction.

Module-1 + Module-2 Integration

Pipeline Connection:

Module-1 (Packet Ingestion & Streaming Generator) streams continuous 46-D telemetry into Module-2 (Adaptive Inference Core & SIEM Dispatcher).

Integration Bugs Met & Resolved:

1. Windows console CP1252 character map crashed on Greek symbols (Delta_H, tau). Resolved with ASCII scientific fallback and UTF-8 stdout wrapper.

2. Scikit-learn 1.7.2 vs 1.8.0 warning spam flooded CI logs with 27 warning blocks. Resolved with targeted InconsistentVersionWarning filters in model_loader.py and pytest.ini, yielding an 84.7% speedup.

My Contribution and Phase-2 Plan

My Individual Contribution

Weeks 1-2 (Dataset Engineering):
Ingested & preprocessed 277,369 balanced flows from the CICIoT2023 benchmark; engineered 46-feature schema across 8 macro attack categories.

Weeks 3-4 (Dual-Tier ML Pipeline):
Built core dual-tier inference engine (src/tier_switcher.py), training Edge Decision Tree and Fog LightGBM models.

Weeks 5-6 (Refinement & Optimization):
Resolved 3 critical production defects, achieved 84.7% test suite speedup, and built real-time edge streaming simulator (cli.py stream).

Role: 100% individual responsibility for pipeline architecture, ML modeling, microservice backend, CLI tooling, and testing harness.

Pending Work (~30%)	Proposed Technical Approach	Target Week
Hardware Gateway Testbed	Flash lightweight ONNX runtime onto Raspberry Pi 4 edge gateway.	Weeks 11-12
Federated Model Updates	Implement secure weight aggregation for distributed fog nodes.	Weeks 12-13
Cloud SIEM Webhook Integration	Connect Kafka / Elasticsearch pipeline for live threat dashboard.	Weeks 13-14

Learnings and Thank You

Hardest Engineering Challenge & Key Learnings

Hardest Problem:
Reconciling the strict sub-millisecond latency requirement of edge IoT nodes (< 1.0 ms) with high multi-class attack classification accuracy (> 99%). Monolithic deep neural nets were too slow (> 80 ms), while standalone lightweight trees suffered from false positives on novel evasion attacks.

What Worked:
Designing a hybrid architecture pairing a sub-millisecond Decision Tree first-tier with an entropy-driven hysteresis controller that escalates only anomalous flows to a Fog LightGBM model.

What I Learnt for Production Systems:
Defensive polymorphic input normalization is non-negotiable in edge services; cross-platform stdout sanitization prevents catastrophic deployment failures; and targeted warning suppression yields dramatic CI/CD speedups.

Thank You. Questions?

Email: raghunatha.reddy@reva.edu.in | GitHub: https://github.com/Raghunath2604/RIO_PROJECT-SMART_CITY- | Console: https://rio-project-smart-city.vercel.app/